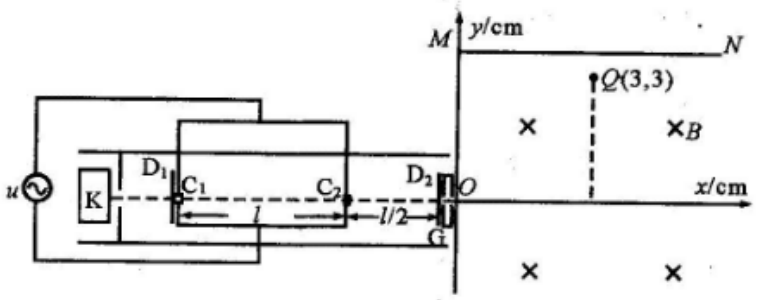
5. (25 points)

K emits positive particles continuously at various speeds, with charge q and mass m, all along the dashed line. D_1、D_2 are timed shutters; C_1、C_2 are gates; C_1 adjoins D_1; gate spacing l. Both gates carry AC voltage $u=U_0\sin(\frac{2\pi}{T}t)$ of period T. Passage occurs only instantaneously near zero voltage. Energy modulator G adjoins D_2, $l/2$ from gate C_2. At time t, G changes energy by $\Delta E_t=E_0\sin(\frac{2\pi}{T}t+\frac{\pi}{4})$, keeping velocity direction. $E_0=\frac{\sqrt{2}}{4}\times m\frac{l^2}{T^2}$. Past G, particles enter at O the uniform field right of G. Its magnitude is B, into the page; all is in vacuum. As shown, origin O defines Cartesian axes Oxy; MN is parallel to the x axis. At $t=0$ open D_1, D_2 together to admit particles to C_1. At $t=3T/4$, close D_1 to stop entry to C_1; at $t=10T/4$, close D_2 to stop entry to G. The fastest entry at O passes Q at (3cm, 3cm). For the slowest entry at O to pass Q, where must boundary MN be shifted parallel to itself in plane Oxy?

五、如图所示，K 为一带电粒子发生器，从中可以不断地射出各种不同速率的带电粒子，它们都带正电，电量为 q，质量为



m，速度的方向都沿图中的虚线. D_1、D_2 为两块挡板，可定时开启和关闭. C_1、C_2 为两扇“门”，C_1 紧靠 D_1，两门之间的距离为 l，两个门上都加上交变电压 $u=U_0\sin\left(\frac{2\pi}{T}t\right)$，$T$ 为交变电压的周期. 已知只有当门上电压的值为零附近的无限短的时间内，粒子才能通过该门. G 为能量增减器，它紧靠挡板 D_2，到门 C_2 的距离为 $l/2$. 当带电粒子在 t 时刻通过G时，粒子可获得一定的能量 $\Delta E_t=E_0\sin\left(\frac{2\pi}{T}t+\frac{\pi}{4}\right)$，但速度的方向不变，式中 $E_0=\frac{\sqrt{2}}{4}\times$